

through fibre optic technology at tera-bits per second speeds, or terrestrial radio-based 4G / 5G wireless telecommunication networks with gigabits per second speeds, even space-based satellite-powered networks capable of megabits per second speeds, one way or another they all connect to the internet via BGP. Despite being such an essential ingredient of the internet's recipe, BGP has some huge red flags.

What's wrong with BGP: How's it a security risk?

In fact, concerns about the fundamental security vulnerabilities in BGP have been there for all to see for "more than 25 years," according to the report by the White House Office of the National Cyber Director. "As initially designed and commonly operating today, BGP does not provide adequate security and resilience features for the risks we currently face," the report goes on to say. What are some of these BGP security risks?



Simply put, BGP doesn't have built-in security features, it operates on implicit trust. BGP's designed for network efficiency – not security in mind. This means BGP can't verify the authenticity of routing information or protect against attacks like route hijacking. It doesn't verify the authority of a network claiming a route, leaving it vulnerable to attacks like route hijacking.

In route hijacking or BGP hijacking attacks, internet traffic is rerouted. Attackers accomplish this by falsely announcing ownership of groups of IP addresses, called IP prefixes, that they do not actually own, control, or route to, according to Cloudflare. A BGP hijack is similar to someone replacing all the road signs on a highway, causing drivers to take wrong exits and get lost, or send

them to unintended hazardous and dangerous destinations.

One of the most recent BGP-related snafu was recorded when Facebook, WhatsApp and Instagram went offline for over 6 hours on October 4, 2021 for 3.5 billion users. The issue was caused by a faulty configuration change that disconnected Facebook's data centres from the internet. Specifically, BGP routes to Facebook's DNS servers were withdrawn, making it impossible for users to resolve Facebook domain names, according to Cloudflare.

The US government strongly supports the FCC's investigation into BGP security risks. In a joint letter dating June 2024, the US Justice and Defense Departments highlighted how China Telecom Americas (CTA) intentionally misdirected American internet traffic to China between 2010 and 2019, exploiting BGP hijacking. This kind of manipulation poses a serious security threat to network security and data integrity, and the FCC ultimately revoked CTA's US operating licence in 2021, according to Reuters.

These are just a couple of the various small and big global incidents where BGP security and routing issues have adversely impacted people's ability to access the internet in a safe and secure manner. So, what's the fix for this glaring vulnerability that underpins our daily digital lives?

Beyond BGP: RPKI and the road to internet safety

Enter Resource Public Key Infrastructure, or RPKI for short. Think of RPKI as a sort of driver's licence system for the internet's routing paths, if we take the earlier example above. It helps verify that the network claiming to route traffic to a particular IP address is indeed authorised to do so. The US government, recognizing the critical nature of BGP's weaknesses, is throwing its weight behind RPKI as a potential industry-wide solution.

The White House's Office of the National Cyber Director (ONCD) is actively calling for widespread adoption of RPKI to shore up BGP's security flaws. By digitally signing BGP route announce-

ments, RPKI can prevent malicious actors from hijacking routes and misdirecting internet traffic. It's like adding a layer of authentication to ensure that when you're sending data to a particular destination, it's actually going to its intended destination.



But, of course, implementing RPKI across the vast, sprawling expanse of the internet isn't as simple as flipping a switch. There are challenges – technical, operational, and even political. For one, not all network operators are on board yet. Smaller internet service providers may lack the resources or expertise to implement RPKI effectively. There's also the matter of global coordination; the internet doesn't respect national borders, and a piecemeal adoption could leave gaps in security.

Moreover, concerns about centralisation and control linger. Some stakeholders worry that RPKI could give undue power to regional internet registries or governments, potentially leading to censorship or overreach. Balancing these concerns with the pressing need for enhanced security is a tightrope walk that requires careful navigation.

Despite these hurdles, the opportunities presented by adopting RPKI are substantial. A more secure BGP means a more resilient internet, less prone to disruptions that can cost businesses millions and erode user trust. It also means thwarting malicious activities like traffic interception, phishing, and denial-of-service attacks that exploit BGP's vulnerabilities. **d**

